

AI Instagram DM → Lead Capture & Qualification Agent

Complete Blueprint & Setup Guide (n8n + AI)

Overview

This system is an **AI-powered lead capture agent** that automatically processes Instagram direct messages, classifies user intent, and stores leads in a structured CRM.

It eliminates manual response handling and ensures **no high-value lead is missed**.

Objective

- Capture incoming Instagram DMs
 - Analyze intent using AI
 - Classify leads into:
 - 🔥 Hot (ready to buy)
 - 🌞 Warm (interested)
 - ❄️ Cold (casual)
 - Store all leads in CRM (Google Sheets)
 - Send instant alerts for hot leads
-

System Architecture

```
graph TD
    A[Instagram DM] --> B[Webhook (n8n)]
    B --> C[Normalize Data]
    C --> D[AI Agent (Intent + Lead Score)]
    D --> E[Merge Data]
    E --> F[Parse Output]
    F --> G[Decision IF Node]
    G -- TRUE (Hot) --> H[Google Sheets + Gmail Alert]
    G -- FALSE --> I[Google Sheets]
```

Core Components

1. Webhook Trigger

- Entry point for incoming messages
- Receives:

```
{
  "username": "string",
  "message": "string",
  "timestamp": "ISO date"
}
```

2. Data Normalization

- Cleans and standardizes incoming data
- Ensures consistent field structure

3. AI Lead Classifier

Uses AI to extract:

- Intent (buying / enquiry / support)
- Lead Score (hot / warm / cold)
- Summary

Output Format:

```
{
  "intent": "buying",
  "lead_score": "hot",
  "summary": "User requests pricing"
}
```

4. Merge Layer

- Combines:
- Original user data
- AI-generated output

5. Parsing Layer

- Converts AI string output into structured JSON
- Ensures compatibility with downstream nodes

6. Decision Engine (IF Node)

Condition:

```
lead_score === "hot"
```

Routes leads into:

- TRUE → High-priority flow
- FALSE → Standard storage

TRUE BRANCH (HOT LEADS)

Actions:

1. Store in Google Sheets

Fields:

- Username
- Message
- Intent
- Lead Score
- Summary
- Status = HOT
- Created At

2. Send Gmail Alert

Email includes:

- Username
- Message
- Intent
- Lead Score
- Timestamp

Purpose: 👉 Instant notification for high-value leads

FALSE BRANCH (WARM / COLD)

Actions:

Store in Google Sheets

Same fields as above, but:

- Status = lead_score (warm/cold)

Purpose: 🙌 Maintain database for follow-ups

Setup Instructions

Step 1: Import Workflow

- Open n8n
 - Import JSON workflow file
 - Verify node connections
-

Step 2: Configure Webhook

- Copy webhook URL
 - Connect it to:
 - ManyChat OR
 - Instagram API OR
 - Manual test tool
-

Step 3: Configure AI Agent

- Add OpenAI API key
 - Use structured prompt for JSON output
 - Disable verbose responses
-

Step 4: Configure Google Sheets

Create sheet with columns:

Username Message Intent Lead Score Summary Status Created At
--

- Connect Google account
 - Map all fields correctly
-

Step 5: Configure Gmail

- Connect Gmail account
 - Add recipient email
 - Customize alert message
-

Step 6: Test Workflow

Test Case 1 (Hot Lead)

Input:

```
{  
  "message": "I want pricing"  
}
```

Expected:

- Stored in sheet
 - Status = HOT
 - Email alert triggered
-

Test Case 2 (Warm Lead)

```
{  
  "message": "Tell me more"  
}
```

Expected:

- Stored in sheet
 - No email
-

Test Case 3 (Cold Lead)

```
{  
  "message": "Hi"  
}
```

Expected:

- Stored in sheet
 - No email
-

Common Issues & Fixes

Issue	Cause	Fix
IF node not working	Formatting issue	Use <code>.trim().toLowerCase()</code>
Missing data	Set node overwrite	Disable "Keep Only Set"
JSON parse error	AI output invalid	Enforce strict JSON prompt
Google Sheets error	Column mismatch	Match exact header names

Future Enhancements

- Auto-reply to DMs
- Lead scoring improvements
- CRM integrations (HubSpot, Airtable)
- Analytics dashboard
- Multi-platform support (WhatsApp, Email)

Use Cases

- Agencies handling Instagram leads
- Freelancers managing inquiries
- SaaS onboarding automation
- Digital marketing pipelines

Outcome

This system enables:

- ⚡ Faster response time
- 📊 Better lead conversion
- 🛠️ Reduced manual effort
- 🔥 Zero missed high-value leads

Support / Access

To get the full workflow or setup help:

- 👉 Reach out or comment **"AI"**

★ Final Note

This is a **scalable foundation** for building advanced AI sales and automation systems.